AUTOMATED GLACIAL LAKE DETECTION USING DEEPLABV3+

Team Information

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Abstract

Glacial lakes are rapidly increasing in number and size due to climate change and glacier retreat. These lakes pose a significant risk because they can trigger Glacial Lake Outburst Floods (GLOFs), causing severe damage to downstream communities and infrastructure. Manual mapping of glacial lakes from satellite imagery is labor-intensive and time-consuming. This project presents an automated deep learning framework for glacial lake detection using DeepLabV3+ with an EfficientNet-B4 backbone. The proposed system performs semantic segmentation on satellite imagery to accurately identify lake regions while minimizing confusion with snow, glacier ice, debris-covered ice, and terrain shadows. The model achieved an Accuracy of 99.62%, Precision of 98.09%, Recall of 93.49%, F1 Score of 95.74%, IoU of 91.82%, and Cohen's Kappa of 0.955.

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1. Introduction

Glacial lakes are water bodies formed by melting glaciers. Due to global warming and accelerated glacier retreat, the number and size of these lakes are increasing worldwide. While glacial lakes are important indicators of climate change, they also present significant hazards. Sudden failure of natural moraine dams can result in Glacial Lake Outburst Floods, leading to catastrophic consequences.

Traditional methods of glacial lake mapping involve manual interpretation of satellite imagery. These methods are not scalable for large datasets and require expert knowledge. Therefore, automated detection systems based on artificial intelligence can significantly improve monitoring efficiency and accuracy.

The objective of this project is to develop an automated framework capable of accurately detecting glacial lakes from satellite imagery.

2. Dataset Description

The dataset provided for the challenge consisted of approximately 575 satellite images containing glacial regions and potential lake formations.

Constraints

- External datasets were not permitted.
- External ground truth labels were prohibited.
- Only challenge-provided imagery could be used.

Challenges Observed

The dataset presented several challenges:

- Snow-covered terrain visually resembles water.
- Glacier ice exhibits spectral characteristics similar to lakes.
- Terrain shadows often appear as dark regions.
- Debris-covered ice creates segmentation ambiguity.
- Significant variation in lake size, shape, and color.

These factors made traditional image processing methods insufficient for reliable segmentation.

3. Data Preparation

Initial Experiments

Initially, several image processing techniques were explored:

- Color Thresholding
- Adaptive Thresholding
- Hybrid Mask Generation

Although these methods generated masks quickly, they produced numerous false positives and failed to generalize across different terrain conditions.

Manual Annotation

To improve label quality, manual polygon annotations were created. During annotation:

- Only confirmed lake regions were labeled.
- Snow regions were excluded.
- Glacier ice was excluded.
- Terrain shadows were excluded.
- Ambiguous regions were intentionally left unlabeled.

This process resulted in a high-quality ground truth dataset suitable for supervised learning.

Data Augmentation

To increase dataset diversity and reduce overfitting, the following augmentations were applied:

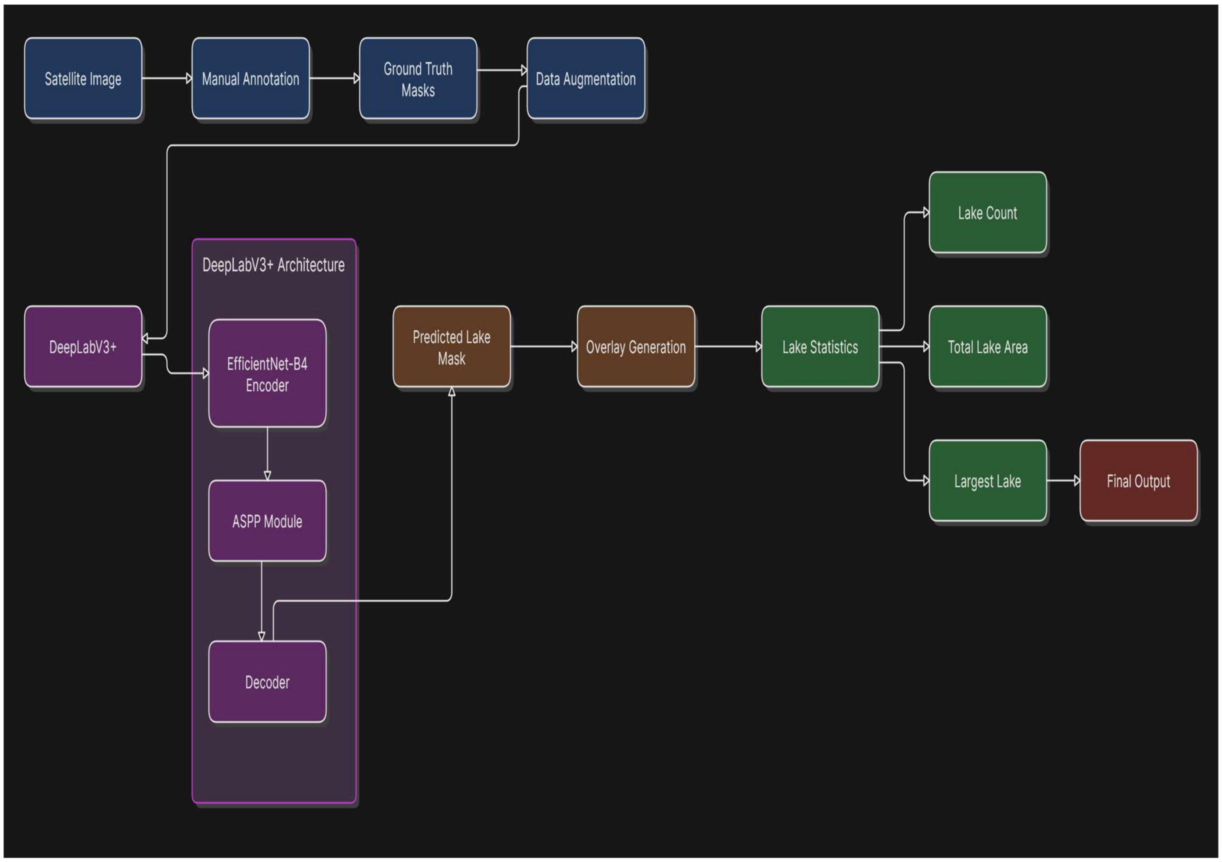
- Horizontal Flip
- Vertical Flip
- Random Rotation
- Brightness Adjustment

- Contrast Adjustment

The final dataset contained 575 image-mask pairs used for training and validation.

4. Proposed Methodology

The proposed methodology follows a complete end-to-end segmentation pipeline.



This framework allows automatic identification of lake regions from previously unseen satellite images.

5. Model Architecture

DeepLabV3+

DeepLabV3+ is a state-of-the-art semantic segmentation architecture designed to capture both local and global contextual information.

Key advantages include:

- Multi-scale feature extraction
- Atrous Spatial Pyramid Pooling (ASPP)
- Accurate boundary delineation
- Strong performance on complex segmentation tasks

EfficientNet-B4 Backbone

EfficientNet-B4 was selected as the encoder due to:

- Efficient parameter utilization
- Strong feature extraction capability
- High accuracy-to-computation ratio

The architecture outputs a pixel-wise segmentation mask indicating lake and non-lake regions.

6. Training Strategy

Image Preprocessing

- Resize: 512 × 512
- Normalization

Optimizer

AdamW Optimizer

Learning Rate Scheduler

Cosine Annealing Learning Rate Scheduler

Epochs

50

Loss Function

To address class imbalance between lake pixels and background pixels, a hybrid loss function was used:

$$\textbf{Final Loss} = 0.5 \times \textbf{BCE Loss} + 0.5 \times \textbf{Tversky Loss}$$

Where:

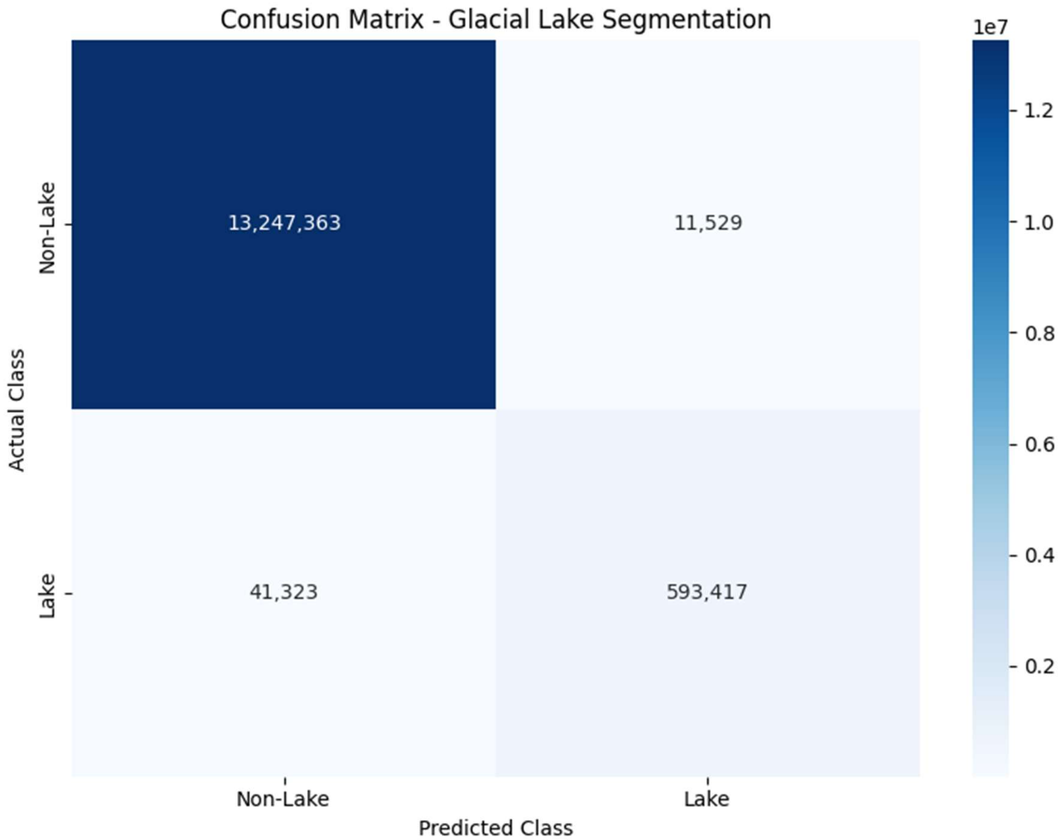
- BCE Loss improves pixel-level classification.
- Tversky Loss improves segmentation performance for minority classes.

7. Experimental Results

The trained model was evaluated on a validation dataset using standard segmentation metrics.

Metric	**Value**
Accuracy	99.62%
Precision	98.09%
Recall	93.49%
F1 Score	95.74%
IoU	91.82%
Cohen's Kappa	0.955

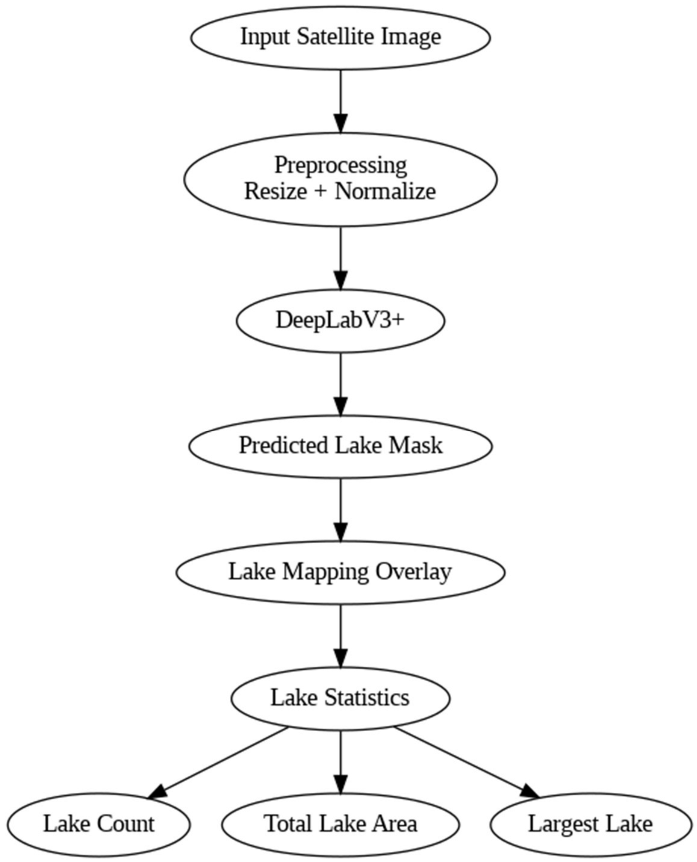
Confusion Matrix



These results demonstrate strong segmentation capability with minimal false positive detections.

8. Inference Pipeline

After training, the model was deployed for inference.



The system automatically generates:

- Binary Segmentation Masks
- Overlay Visualizations
- Lake Count
- Total Lake Area
- Largest Lake Region

The model was successfully tested on unseen satellite imagery, including Kodaikanal Lake, demonstrating strong generalization capability.

9. Applications

Potential applications include:

- Glacial Lake Monitoring
- GLOF Risk Assessment
- Climate Change Studies
- Remote Sensing Automation
- Environmental Monitoring

The proposed framework can significantly reduce the effort required for manual lake mapping.

10. Conclusion and Future Work

This project presented an automated glacial lake detection framework based on DeepLabV3+ and EfficientNet-B4.

Major achievements include:

- Creation of high-quality manual annotations.
- Development of a robust segmentation pipeline.
- Achievement of IoU = 91.82%.
- Achievement of F1 Score = 95.74%.
- Successful detection of lakes in unseen imagery.

Future work may include:

- Additional manual annotations.
- Test-Time Augmentation.
- Multispectral Sentinel-2 analysis.
- Digital Elevation Model integration.
- Automated GLOF risk estimation.

The proposed framework provides a scalable and accurate solution for automated glacial lake monitoring and hazard assessment.